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SDP Proposal

Design and Implementation of an Edge-Based Driver Monitoring System with False Alarm Reduction

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Abstract

In-cabin Driver Monitoring Systems (DMS) play an increasingly vital role in road safety, particularly as regulatory bodies such as the European Union (GSR 2024) mandate automated alertness and distraction detection in all newly registered vehicles. However, conventional commercial and edge-deployed systems suffer from elevated false alarm rates, triggering nuisance alerts that induce driver frustration, cognitive fatigue, and system deactivation. This Senior Design Project presents a reliable, edge-based Driver Monitoring System deployed on the Raspberry Pi 5 platform paired with a 1080p 30 FPS RGB camera. The architecture integrates a fine-tuned YOLO11n object detector with hard-negative mining, casual exponential moving average (EMA) temporal filtering, mirror-check suppression, camera-blindness detection, and tiered multi-modal alerts in English and Arabic. Furthermore, privacy-preserving event logging with power-cut-safe SQLite WAL storage synchronizes with an interactive desktop and tablet dashboard providing behavioral trends, safety KPIs, and intelligent rest-stop suggestions.

Chapter 1 Introduction

1.1 Problem Statement and Proposed Solution

Driver Monitoring Systems (DMSs) evolved since they got introduced in 2006 Lexus and Corporation [2006]. Transitioning from merely alerting the driver through an Infrared (IR) camera and applying breaks Smith et al. [2012], to tracking eye gaze and fusing it with vehicular spatial awareness Zhang et al. [2020]. Conversely, nuisance alerts and false alarms causes driver distrust in these systems, causing system neglect or circumvention Parasuraman and Riley [1997]. The gap falls under improving the user experience in theses systems. Given the United Arab Emirates (UAE) context of high speed highways, and the need for private and trustworthy systems. We aim to deliver a system that is built the supplement the driver's attention and not replace it. Figure 1.1 shows the results of in-attentive driving, which stresses the need for these systems to be deployed at a greater rate. In 2024, European Union (EU)'s General Safety Regulation (GSR) mandates that all newly registered vehicles must include a DMS European Parliament and Council of the European Union [2019]. To supplement this surging rate of deployment, the need for systems that allows the user to entrust them becomes apparent. We aim to achieve this by adaptive and enhanced alerting, driving analytics and a reliable drowsiness and distraction monitoring system.

1.2 Limitations of Traditional Systems

DMS is notorious for the high rate of alerts, a study argues that frequent or overlapping alerts can negatively affect driver mental models.Furthermore, the authors stress a need for improved contextual calibration in future driver monitoring systems designs Bainbridge and Stanton [2020]. Figure 1.2 highlights a local survey where drivers relay the common causes of distraction. This exposes a common pitfall that traditional DMS suffers from. By only tracking the driver's face and eyes, these systems cannot detect external distractions like road signs and traffic, nor can they detect noise and mental distractions caused by passengers or car controls. Later in this report, we discuss further details on how we effectively counter the

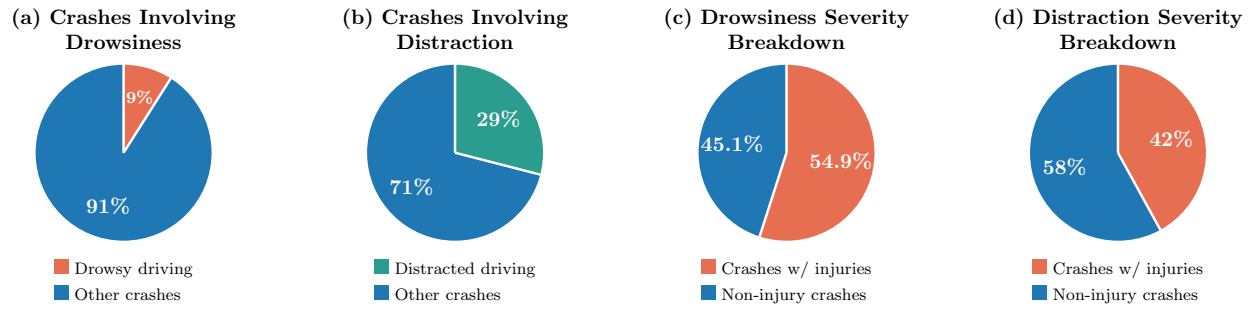


Figure 1.1: (a) Drowsy driving involvement in all crashes National Highway Traffic Safety Administration (NHTSA) [2021]; (b) Distracted driving involvement in all crashes National Highway Traffic Safety Administration (NHTSA) [2022]; (c) Severity of drowsiness-caused crashes Tefft [2016]; (d) Severity of distraction-caused crashes Dingus et al. [2016]

aforementioned shortcomings, using initial setup driver calibration, a user-friendly system configuration device, and context-aware alerting.

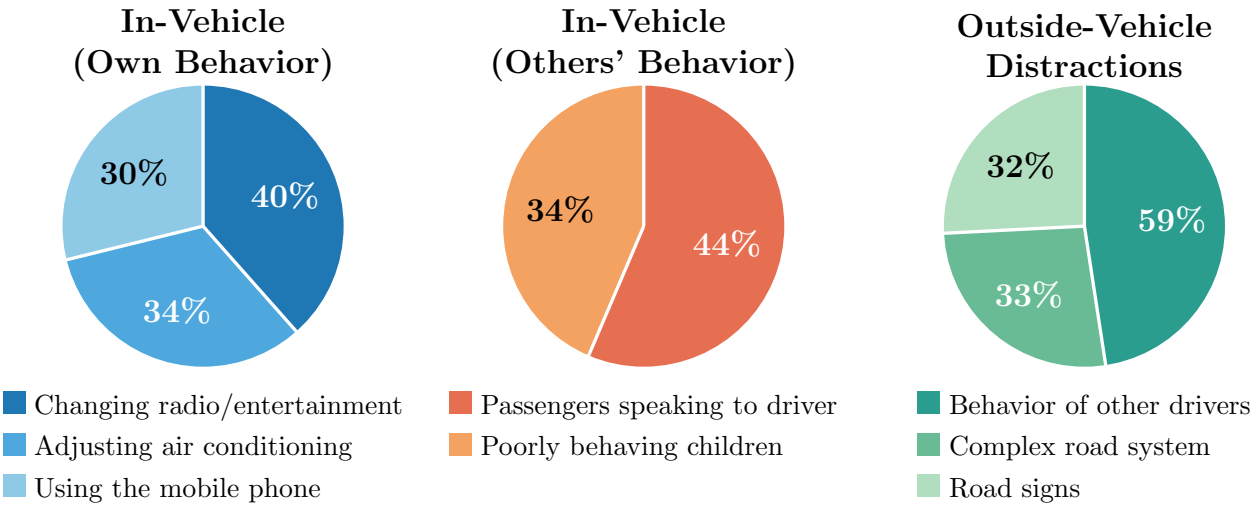


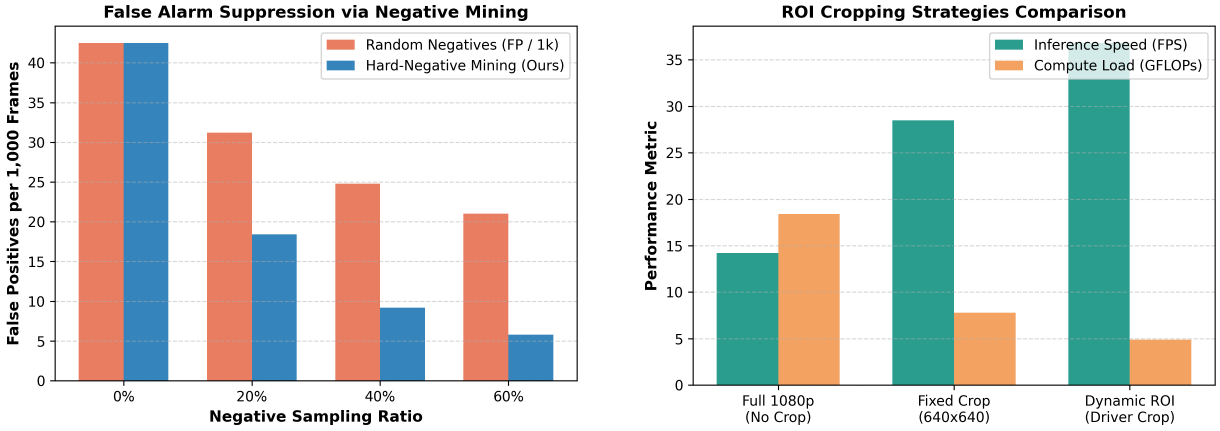
Figure 1.2: (a-c) Local survey on driver distraction causes RoadSafetyUAE and YouGov [2022]

1.3 Key Differentiators

Our solution has several clear differentiators from previously deployed systems. The overall system is built on a lightweight You Only Look Once (YOLO) object detector that is further pruned and quantized to run at a streamlined latency with high Frames Per Second (FPS) throughput. The model is initially pre-trained via Common Objects in Context (COCO). Then, fine-tuned via our own subject-disjoint in-cabin Red-Green-Blue (RGB) video dataset,

which involves ranging lighting conditions to train a model where it is most critical. Later, we extend the dataset to include harder negatives to help reduce false alarms. We pair this detection model with a casual temporal filter to reduce random detection spikes and improve user experience via escalating alarms based on risk assessment. The system extends these with a privacy-first logging system where only event records are transmitted via Transport Layer Security (TLS) to a server database where a web dashboard relays driving analytics, giving feedback via tips and behavior trends to the driver after the session. A driver-facing easy-to-navigate tablet allows for system calibration and initial setup. We pair traditional alerts with intelligent voice messages in English and Arabic for further transparency and rest-stop location suggestions via a Global Positioning System (GPS).

Aside from these differentiators, the team initially conducted research via empirical methods to evaluate several object detection models, and the effect of hard-negative mining on false alarms. We later conducted ablation studies on temporal filters and the effect of ranging input resolution and Region of Interest (ROI) cropping on input video streams, which led to most of our design choices in this report to be backed by real evaluations.



(a) Hard-negative mining

(b) ROI cropping methods

Figure 1.3: Empirical evaluation conducted in this work (**Ours**):
(a) Suppression via hard-negatives ; and (b) Evaluating ROI cropping strategies.

1.3.1 Target Users and Markets

We aim to provide this trustworthy solution to everyone. Primarily, our solution serves both the general public and commercial vehicle fleet operators. Specifically, government agencies such as Dubai's Roads and Transport Authority (RTA), which deploys thousands of vehicles and employs personnel who drive for extended periods (such as taxi drivers), would benefit

from a private, dependable safety system to support these operators.

1.4 Scope and Feature Priorities

Features are grouped by priority, "Required" features that deliver the base build, "Desired" features that will be delivered depending on the remaining time budget, and "Future" features that are out of scope of our Senior Design Project (SDP) 2. In the following sections we list the breakdown in detail.

Table 1.1: Required System Feature Specifications

Status	**Title**	**Description**
Required	Deployment on Raspberry Pi 5	The full detection pipeline runs live on the Pi 5.
Required	1080p 30 FPS RGB camera	A Universal Serial Bus (USB) 3.0 webcam aimed at the driver's face records 1080p at 30 FPS. Each frame is cropped to a fixed region before detection.
Required	Object detection model	YOLO11n detects drowsy and distraction classes. It is fine-tuned on the subject-disjoint dataset below.
Required	Temporal filtering (Exponential Moving Average (EMA))	An exponential moving average over each class score suppresses random single-frame spikes. It is defined in seconds, not frames, so it behaves the same at any FPS.
Required	Speaker and Light Emitting Diode (LED) bar alert	After a confirmed detection, a bright LED bar flashes and a USB 3.0 speaker plays a voice warning.
Required	Alert cool-down	Cheap and effective suppression for repeated warnings for the same event within a time frame.
Required	Mirror-check suppression	The classes (by design) do not depend on head orientation, so normal mirror checks and head turns do not trigger alerts.
Required	Internal logging	Each confirmed detection (time, type, confidence) is saved to the Pi's local storage. No raw video is kept by default.
Required	On-device video processing	All video processing happens on the Pi and raw video never leaves the vehicle.
Required	Subject-disjoint dataset collection	10-minute RGB in-cabin videos over several sessions, recorded in the morning and evening. Subjects are split so no subject appears in more than one of train, val, and test.

Table 1.2: Desired System Feature Specifications

Status	**Title**	**Description**
Desired	Model pruning and quantization	Pruning removes channels to cut Floating Point Operations (FLOPs). Quantization methods are compared on the Pi, choosing the best FPS while retaining detection metrics.
Desired	Hard-negative dataset	Subjects are re-recorded performing look-alike non-violations flagged by the baseline model. Only negative frames are added.
Desired	Neutral baseline clips	Each subject records a 20 s neutral baseline clip to benchmark and evaluate calibration.
Desired	Class-specific alerting	Distinct alert tiers and voice notifications per class.
Desired	Tiered alert escalation	Gentle notification $\rightarrow$ voice prompt $\rightarrow$ multi-modal alarm (LED bar and voice) as risk persists.
Desired	Clear, explainable alerts	Contextual alerts (e.g., "yawning detected 3 times in the last 5 minutes") rather than generic warnings to build driver trust.
Desired	Camera-blind check	Measures frame brightness and sharpness. Logs and issues "camera blocked" warning if unclear for ≥ 3 s.
Desired	Driver-centered crop tracking	Lightweight face detector runs at ~ 1 Hz on lower resolution frames to dynamically adjust the face crop.
Desired	New-driver calibration	Driver sits neutrally for 20 s while detector runs at low confidence. Cutoffs set to 99th percentile + a safe gap. Increasing the confidence threshold and reducing false alarms
Desired	Eye closure and micro-sleep alert	Face landmark tracking calculates Eye Aspect Ratio (EAR). Continuous closure triggers loud alert; 60 s Percentage of Eye Closure (PERCLOS) triggers gentle fatigue prompt.
Desired	Power-cut-safe logging	Database operates in Write-Ahead Logging (WAL) mode to prevent corruption during power interruptions.
Desired	Dashboard and cloud sync	Post-trip event records uploaded over TLS to update database and web dashboard with trends and metrics.
Desired	Actionable post-trip tips	Recommendations generated from logged behaviors (e.g., "take a 30 minute break")
Desired	Driver-facing tablet	Dedicated low-distraction glance-able User Interface (UI), allows for easy system calibration.
Desired	Pre-trip readiness check	Pre-drive fatigue survey on tablet recommending rest and nearby rest locations if fatigue is reported.
Desired	Rest-stop suggestions GPS	Queries GPS and Point of Interest (POI) map services to recommend nearby rest stops or fuel stations when violations exceed a threshold.

Table 1.3: Future System Feature Specifications

Status	Title	Description
Future	Driving-context suppression	Suppress or defer warnings during reversing maneuvers, stand-still red lights, or high-density traffic.
Future	Multi-modal fusion	Aggregate multi-spectral video streams.
Future	Multiple camera feed	Additional camera to capture wide-angle context or secondary angles.
Future	Smartwatch and wearables integration	Incorporate driver heart-rate or biometric motion telemetry to augment edge detection.
Future	Robustness study	Systematic stress testing across glare, low light, optical blur, pose deviations, eyewear (glasses/sunglasses), face coverings, day-to-night transfer, and multi-vehicle cabins.
Future	Adaptive sensitivity by context	Context-aware threshold modification depending on the driver's behavior patterns.
Future	Steering wheel vibration	Haptic vibration on the steering wheel to reduce driver reaction latency.
Future	Cabin comfort integration	System can change the Air Conditioning (AC) or change the volume of what's playing via the car's speakers.
Future	Voice recognition	Driver audio-interface enabling inquiries (e.g., "How tired am I?").

1.5 Proposal Overview and Revisions

Our initial proposal covered the pre-development in detail, discussing a methodology for evaluating different object detectors and studying the effect of hard-negative mining. The primary focus was edge system performance on accuracy metrics and False Positives (FPs) per 1000 frames. Since then, we changed the deployment hardware, upgrading from a Raspberry Pi 4 to a Pi 5 to provide better throughput and handle larger systems. We then pivoted the project to provide better User Experience (UX) through various adaptive alerting mechanisms, nuisance filters, and system components such as web analytics or a user-friendly tablet interface.

1.6 Project Roadmap

We studied and evaluated different object detectors on RGB and IR public DMS datasets. Followed by evaluating negative-mining on false detections, and an empirical study on the trade-offs between temporal filters. Finally, we studied the effect of pruning, ROI cropping and input resolutions on performance.

We will establish our organic in-cabin video dataset, with an extension of harder non-violation clips that will harden our detection model. Delivering a public subject-disjoint

dataset.

Before deployment, we perform an extensive literature review, to find what do drivers suffer from by using traditional DMS. Aiming to effectively counter these pain-points through careful system planning and design. we plan to document every design change meticulously. If the system faces technical complexity issues, we plan to surgically remove any non-required component, ensuring project delivery on time.

Post-deployment, we evaluate the system by involving users for their feedback, adequate testing and taking recommendations from peers, allowing the project to develop and adapt to fit the end user the best.

1.7 Report Organization

To end this chapter, here is a breakdown on how this report is organized:

In Chapter II, we explore current limitations and shortcomings in existing DMS work, providing a gap summary highlighting the need for better UX in said systems.

In Chapter III, we deep-dive into the technical concepts and principles of deploying complex DMSs.

In Chapter IV, we report in detail our proposed solution, funneling a general idea into a detailed look at every component in the system. To provide a complete report, project constraints and issues are discussed.

Chapter V includes the methodology, highlighting the empirical methods we used to evaluate, organize and build the system and its dataset.

Finally Chapter VI summarizes the main findings, lists the project milestones and discusses what we aim to work on in SDP 2.

Chapter 2 Literature Review

This chapter explores the rapid growth in the utilization of Unmanned Aerial Vehicles (UAVs) and the associated risks, particularly as UAVs become increasingly integrated into civil, commercial, and defense sectors. Their transformative impact on air operations is largely attributed to advancements in navigation, sensing technologies, and miniature satellites. Today, UAVs are deployed in diverse applications such as agriculture, logistics, environmental monitoring, and infrastructure inspection. However, despite their capabilities, UAVs pose significant challenges, particularly in airspace management—specifically regarding collision avoidance, traffic coordination, and adherence to no fly zones.

Chapter 3 Technical Background

Chapter 4 Proposed Solution

Chapter 5 Research Methodology

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